Ali Can Aydoğdu

Process Development & Production Manager · Mechanical Engineer

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SUMMARY

I am a results-driven Mechanical Engineer with a strong 11-year career in the automotive industry. My expertise is built on 8 years of process optimisation / production management and 3 years of custom machine design experience. Along my career journey I have developed a broad, complementary skill set, from strategic investment planning to operational management on the shop floor. While running the RFQ-to-SOP processes of complex projects end to end — injection machines and equipment, robotic automation lines and Solar Power Plants (GES) — I also led production teams and made sure these operations stayed sustainable.

EXPERIENCE

Industrial Process Expert

2025 – Present

Tredin — Diniz Holding & Treves Group · Bursa, Türkiye

- Benchmark studies across the processes and machine parks of group companies, rolling out industrialization activities and standards
- Running industrialization activities, generating and implementing productivity projects
- Resolving chronic production and measurement errors
- Led projects for the in-house fabrication and commissioning of critical equipment.
- Delivered cost savings compared to outsourcing and improved project lead times.

Feasibility · Machine fabrication · Commissioning · Cost reduction · PFMEA · RFQ · 8D · Productivity Projects · Kaizen

Project codes: K1320 · P1310 · V769 · BC3 · V363 · F2U · F2X

Process Development and Production Chief

2020 – 2024

ORAU Orhan Automotive — Orhan Holding · Bursa, Türkiye

Dual responsibility for managing both the Process Development and Production departments in a high-volume automotive plastics manufacturing plant.

- Led all operational activities for a team of 25 blue-collar and 2 white-collar employees and a machinery park of 30+ units (pipe/coating extrusion lines, planetary machines, injection moulding machines), overseeing performance, motivation and training plans.
- Drove process improvement initiatives by analysing performance losses: a 4% increase in OEE and a 15% reduction in scrap rates.
- Implemented corrective actions to meet strict quality targets and customer standards, successfully reducing customer complaints.
- Minimised planned downtime by 50% through rapid troubleshooting and proactive machinery improvement projects.
- Prepared and managed the annual €** CAPEX budget for all Common Machinery (injection & extrusion), aligning investments with the master plan and new project RFQs.
- Managed the end-to-end feasibility, RFQ and commissioning of a €** Robotic Automated Plastic Production Line.
- As an energy committee member, led the feasibility study and project management for the factory's Solar Power Plant (GES): a 1,200 kWp system commissioned.
- Managed complex, non-standard projects: new plant installations, full commissioning of a plastics factory including infrastructure works, and machine line research and commissioning for new products such as WIT injection and armoured steel cable.

OEE +4% · Scrap -15% · CAPEX €** · Robotic line €** · Solar 1,200 kWp · 8D · Team Management · Leadership · Communication Skills · Executive Reporting

Project codes: CMP · EMP2 · BJA · 356 · HJD · LFF

Process Development Engineer

2018 – 2020

ORAU Orhan Automotive — Orhan Holding · Bursa · Türkiye, Mexico, Morocco

Commissioning activities for machines within the Common Machinery scope.

- Performed commissioning activities for Common Machinery machines across Orau sites (Türkiye, Mexico, Morocco).
- For sub-processes, carried out work from RFQ through to handover to serial production: PFMEA, synoptic, layout, risk analysis, operation instructions, etc. (vertical/horizontal injection, zamak, extrusion, sliding-head lathe).
- Led the WCM Energy Pillar.

Commissioning · RFQ · PFMEA · WCM

Design and Manufacturing Engineer

2015 – 2018

Cisim Mühendislik · Bursa, Türkiye

Design, manufacturing and commissioning of custom machinery and automation projects.

- Acı Pars (Iran) — Servo press conveyor line (disc assembly).
- Franke (Manisa) — Assembly lines (hood production line) and an automatic welding machine.
- Bosch (Bursa & Manisa) — Test machines and apparatus.
- Tofaş (Bursa) — AGV (automated guided vehicle) and welding fixtures.
- Vaillant Demir Döküm — Orbital copper pipe cutting and calibrating machine.

SolidWorks · AutoCAD · AGV · Welding line

ROADMAP

2015: Started my career at Cisim Mühendislik as a Design & Manufacturing Engineer, designing custom machinery.

2016: Moved from design into build; gained on-site machine installation and commissioning experience.

2017: Ran custom machine projects end to end, from design through to commissioning.

2018: Joined ORAU and moved into process engineering; carried out RFQ and PFMEA studies for sub-processes.

2019: Deepened my process optimisation and measurement systems work, applying quality tools (8D, MSA, SPC) on the shop floor.

2020: Became Process Development & Production Supervisor, taking charge of 25 blue-collar + 2 white-collar employees and a 30+ machine park.

2021: Turned process improvement projects into measurable results: higher OEE and lower scrap.

2022: Moved into the investment side, owning feasibility, CAPEX planning and commissioning.

2023: Started my business master's (e-MBA), adding financial literacy and a cost/budget perspective.

2024: Added sustainability and energy management to my portfolio through the Energy Committee and the solar plant project.

2025: Moved to Tredin & Treves Group, running industrialization processes.

2026: Today, with eleven years of accumulated experience, I approach things holistically: I take a problem up together with its process, investment and team dimensions and build lasting solutions.

2028+ (goal): To carry the production, investment and team management muscles I have built into macro-level areas of responsibility where I can directly shape strategic decision-making, producing the highest operational efficiency.

EDUCATION

Süleyman Demirel University

2010 – 2014

Mechanical Engineering · Isparta, Türkiye

- TÜBİTAK-funded graduation thesis (undergraduate final project).

Ahmet Yesevi University

2023 – 2024

e-MBA (distance learning) · —

- The business master's gave me financial literacy, cost and budget management, investment feasibility and project economics — the ability to combine an engineering view with business results.

SKILLS & TOOLS

Management & Leadership: Team management (25 blue-collar + 2 white-collar) (Advanced), Production & operations management (Advanced), Multi-project & portfolio management (Advanced), Performance & KPI tracking (Intermediate–Advanced), Budget & CAPEX management (Intermediate–Advanced), Supplier & RFQ management (Intermediate–Advanced), Stakeholder & executive communication (Intermediate–Advanced)

Process & Quality: Process development (Advanced), Kaizen · Poka-Yoke · 5S (Advanced), Process design (Intermediate–Advanced), Cost reduction (Intermediate–Advanced), Quality (8D, MSA, SPC) (Intermediate–Advanced), PFMEA (Intermediate–Advanced)

Investment & Feasibility: Feasibility & CAPEX planning (Advanced), Cost-benefit & payback analysis (Advanced), Machine/equipment selection & RFQ (Advanced), Commissioning management (Advanced)

Energy & Sustainability: Energy loss analysis (Advanced), Energy project portfolio (60+ projects) (Advanced), WCM Energy Pillar (Intermediate–Advanced), Solar plant feasibility & project management (Intermediate–Advanced)

Automation & Digitalisation: Robotic line installation & integration (Intermediate–Advanced), Data collection & reporting (OPCUA) (Intermediate–Advanced), Mobile production tracking dashboard (Intermediate), Production software (SAP, E-Kanban) (Intermediate)

Design & Manufacturing: Machine & fixture design (Advanced), Technical drawing and layout planning (Intermediate–Advanced), Custom machine build follow-up (Intermediate–Advanced), Prototyping (Intermediate–Advanced)

Personal Competences: Analytical thinking (Advanced), Complex problem solving (Advanced), Conflict management (Intermediate)

Tools: MS Office, SolidWorks, AutoCAD, Node-RED, Android Studio, 3D Printing

PORTFOLIO

Agile Project Management

Scrum Master — multidisciplinary project team

After completing our in-house Agile project management training, I took on the Scrum Master role in a multidisciplinary project team drawn from different locations and departments, and we delivered two major projects in three-month cycles. Working with team members from various disciplines, each with their own KPIs and perspectives, broadened my vision considerably. Covering both Scrum Master and 'Voice' duties, I took my ability to unite the team around a shared vision, find common ground and steer project dynamics with agility to a higher level.

`Energy Efficiency · MAS+ production module`

Robotic Automated Plastic Production Line

2020–2024

Project manager — feasibility, RFQ, commissioning

Delivered a €** robotic automation line end to end, moving manual stations to automation. Throughout commissioning I worked in close contact with the product, laboratory and design departments; sample approvals, test results and design revisions ran through a single coordination plan, and I handed the line over.

`Robotic automation · €** · Commissioning`

100% Visual Inspection Applications in Plastics

2020–2024

Machine and equipment integration — industrial robot + camera system

Visual and dimensional inspection units built with industrial robots and camera systems integrated into the plastic injection machine; machines and equipment for automatic feeding with vibratory and magazine systems, fault handling and 100% inspection.

`100% inspection · Industrial robot · Camera system · Automatic feeding`

Production Management

2020–2024

Process development and production chief — automotive

Capacity planning, line balancing, standard work practices, daily production performance meetings and the development of downtime management processes. SMED studies to reduce OEE losses, improvement of mould changeover times, coordination of maintenance activities, and strengthening shop-floor discipline and operational standards through 5S. Within quality assurance, developing Poka-Yoke solutions, PFMEA and control plans, and managing root-cause analyses and permanent corrective actions with the 8D methodology.

`OEE · Kaizen & SMED · TPM / 5S · Poka-Yoke · Team management · Shop-floor training`

SAP and E-Kanban Integration

2020–2024

Production management — system and shop-floor integration

SAP and e-Kanban went live at the same time; there was no SAP beforehand — material, BOM and production data were managed in a different software. The real difficulty came during the transition: carrying the legacy system's data into the new structure, keeping stock accuracy and production planning running in parallel throughout the migration, and making sure material flow was never interrupted while both systems ran side by side. That period gave me the experience of making fast decisions and adapting quickly on stock and production planning, standardising data and getting the shop floor used to the new flow; the result was a flow tracked through stock accuracy, kanban turns and on-time delivery indicators.

`SAP (MM/PP) · E-Kanban · Transition management · Stock & production planning`

Solar Power Plant (GES) — 1,200 kWp

2020–2024

Energy committee member · project management

Feasibility study and project management for the factory's solar power plant: 1,200 kWp commissioned, projected to save €** annually.

`Energy efficiency · 1,200 kWp · €**/year`

€** CAPEX Budget Management

2020–2024

Common Machinery investment plan (injection & extrusion)

Prepared and managed the annual €** CAPEX budget, aligning investments with the master plan and new project RFQs.

`CAPEX · €** · Master plan`

New Plant Installation & Full Commissioning

2020–2024

Project manager

Full commissioning of a plastics factory including infrastructure works; machine line research and installation for new products (WIT injection, armoured steel cable).

`New plant · Infrastructure · WIT injection`

Tofaş — AGV (Automated Guided Vehicle)

2015–2018

Design, manufacturing and commissioning

Design, manufacturing and on-site commissioning of the AGV project for Tofaş at Cisim Mühendislik.

`AGV · Tofaş · Automation`

Bosch — Test Machines & Apparatus

2015–2018

Design and manufacturing

Design, manufacturing and commissioning of test machines and apparatus for Bosch Bursa and Manisa plants.

`Test machine · Apparatus · Bosch`

International Assignments — Locations

2015–2024

Iran · Tehran — Mexico · Sabinas — Morocco · Tangier

Experience gained on international assignments: working with different cultures, communicating in multicultural teams and coordinating technically with local suppliers; project meetings run in English, on-site installation and commissioning. Managing cultural differences, adapting quickly to different ways of working and delivering projects on time.

International assignment · Cross-cultural communication · Installation & commissioning

Plastics Optimisations

2018–2024

- Mould plate standardisation
- Quick hydraulic clamping
- Pick & place robot applications
- Heat jacket (barrel insulation)
- Central raw-material system
- Mould–machine matching optimisation
- Granulators
- Servo-hydraulic conversions
- Hot runner
- OPCUA data management

PP · TPE · HDPE · POM · PA66 · PTFE · PBT

ENERGY

Energy Committee Member · WCM Energy Pillar

2020–2024 · ORAU Orhan Otomotiv — Bursa, Gemlik, Karacabey

I analysed the plants' energy losses down to department and machine level, ranked improvement projects by annual saving, cost and benefit/cost ratio, commissioned them with the shop-floor team and reported the results. In the same period I ran the Solar Power Plant (GES) project.

- Loss analysis: broke consumption down by department and machine (injection, extrusion, compressors, lighting, offices) and classified loss types.
- Project pipeline: compared 60+ energy projects on annual saving, cost and benefit/cost ratio and set the investment priority.
- Commissioning: implemented the projects and verified them by tracking forecast vs. realised saving and cost.
- Roll-out: took solutions proven on a pilot line to the other lines and plants (Gemlik, Karacabey).
- Reporting: ran the monthly table of electricity consumption (kWh), TEP, CO2 (tonnes), revenue and energy intensity (kWh/TRY).

Electricity consumption: ≈4.4 GWh · TEP: ≈380 · CO2: ≈2,000 t · Electricity cost: ≈€** · Energy intensity: ≈0.16 kWh/TRY

Compressor air jet → blower conversion — ≈€**/yr · ≈€** / line. Replacing compressed-air jets with blowers on extrusion lines (≈65 MWh/yr, B/C ≈ 15). After the pilot line, roll-out to 7 lines: ≈€** investment → ≈€**/yr and ≈500 MWh saved.

Heat jacket application (injection) — ≈€**/yr · ≈€** / machine. Heat jackets on injection machines — 650 t FANUC first, then 300/320/220 t presses; direct saving on heating energy.

Hydraulic pump → servo pump retrofit — ≈TRY 290k/yr · ≈TRY 80k. 300 t Krauss injection machine. Payback ≈ 3.3 months; the highest-return single-machine investment in injection.

Compressor heat recovery system — ≈TRY 825k/yr · Feasibility stage. Recovering waste heat from the LKV 110 kW compressor — the highest saving forecast in the project pipeline.

R&D office LED retrofit — ≈€**/yr · ≈€**. 251 luminaires, 110 W → 42 W (≈29 MWh/yr saved). Payback ≈ 7 months.

IE2 → IE4 motor retrofit & office thermostats — ≈€**/yr per motor · Phased. Replacing 1.1–7.5 kW motors with high-efficiency units and adding thermostats to open-office air conditioners (≈TRY 17k/yr saving).

Solar Power Plant (GES) — 1,200 kWp

- Feasibility and project management: 1,200 kWp installed capacity; projected saving of ~€** per year.
- Generation tracking: monthly target table (≈84 MWh in January → ≈189 MWh in July) plus a daily output vs. target ratio.
- Deviation analysis: output of ≈77 MWh against a ≈108 MWh target in the monitored period; the gap and its cost impact (≈TRY 68.5k) were reported.

FAIRS & SEMINARS

- K Fair Düsseldorf (Germany) — the world's leading plastics and rubber fair; I reviewed injection and extrusion technologies and met machine, mould and automation suppliers on site.
- TÜYAP Maktek Fair (İstanbul) — machine tools and metal-working technologies.
- Tube & Steel Fair — tube, profile and steel processing technologies.
- Bursa Automation Fair — industrial automation, robotics and control systems.
- Automation and Machine Technologies Fair — WIN EURASIA.
- GES seminar — solar power plant technologies and field applications.

INTERESTS & LANGUAGES

Node-RED — Visual programming for home and data automation tasks. · 3D Printing — Prototyping and practical fixes for broken parts. · Camping — Clearing my head outdoors, away from technical work. · AI & AI Agents — Building AI agents for my own workflows: LLM-driven automation, local model experiments and small tooling.

Turkish: Native · English: B2